The Fertility, Marriage, and Gender Equality Quandary

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October 2026

Motivation

- Three demographic transformations over the past half-century:
  1. Falling fertility, often well below replacement
  2. Declining dual parenthood: rising single-parent households
  3. Converging gender income gaps (Goldin 2014)

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  2. Pro-marriage: children's outcomes (Autor et al. 2019)
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- These agendas are discussed separately—yet the underlying decisions are closely connected

This Paper

1. Document a three-way trade-off in 115 countries; same magnitude in OECD & non-OECD but different dominant tensions
2. Develop an equilibrium marriage-market model: marriage bundles income transfer with joint fertility
3. While level policies improve two at the expense of the third
   (a) coordinated instruments can improve all three
   (b) childcare redistribution as a single-instrument alternative
4. Calibrate to Mexico 1990–2025; scope for further redistribution widens as wages converge, but fell short of equal sharing

Related Literature

- Fertility & female labor supply through time cost of children (Erosa et al. 2016, Blundell et al. 2016; Feyrer et al. 2008)
  → Place inside marriage market where the marital contract adjusts
- Marriage-market models (Choo & Siow 2006, Chiappori et al. 2017)
  → Marriage bundles income sharing with joint fertility
- Gender revolution: men's domestic roles restabilize families (Goldscheider et al. 2015, Doepke & Kindermann 2019)
  → Identify bilateral conditions under which this conjecture holds
- Technology & the family (Greenwood et al. 2005, Ngai & Petrongolo 2017)
  → Decompose into gender-neutral and gender-biased components

Motivating Facts

Data Construction

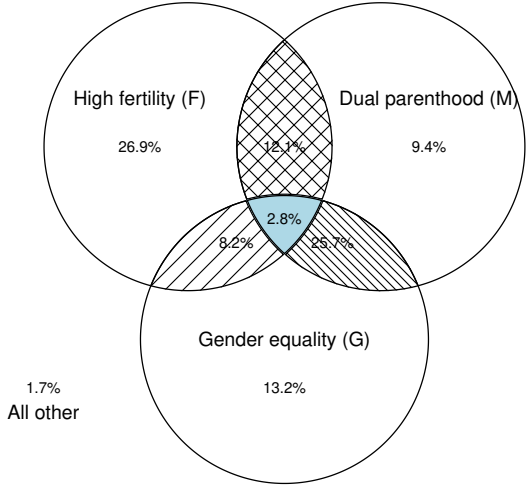
- Fertility: total fertility rate (WDI)
- Dual parenthood: share of children living with both parents
- Gender equality: female share of labor income (WID)

Sample	Countries	Obs	Dual parenthood source
Full	115	1,764	IPUMSI + OECD Family Database
OECD	39	879	OECD Family Database (+ IPUMSI)
Non-OECD	76	885	IPUMSI only

- Construct dummies at sample-specific medians: high F, dual M, equal G
- Dartboard benchmark: under independence, trinity share $\approx 0.5^3 = 12.5\%$

The Trinity is Rare: Full Sample

bootstrap



Same Shortfall, Different Sources

Venn diagrams bootstrap

- Divide into non-OECD and OECD countries

$$\underbrace{p_{111} - p_F p_M p_G}_{\text{shortfall}} = \underbrace{\delta_{FM} \cdot p_G}_{\text{F–M}} + \underbrace{\delta_{FG} \cdot p_M}_{\text{F–G}} + \underbrace{\delta_{MG} \cdot p_F}_{\text{M–G}} + \varepsilon$$

- δ_{XY}: excess joint attainment of pair X, Y vs. independence
- p_X: marginal attainment rate of X

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	F–M	F–G	M–G	Residual	Shortfall
Non-OECD	26.7%	55.9%	37.9%	-20.5%	-6.38 pp
OECD	63.4%	-1.1%	45.1%	-7.4%	-6.34 pp

- Bottleneck: G in non-OECD vs. M in OECD

Questions for Theory

1. What explains the tension?
2. Why do the dominant tensions differ across OECD and non-OECD countries?
3. Can policies address the three-way trade-off?
4. What drives the demographic transition quantitatively?

Theory

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{♂, ♀\}$ and preference

$$u(c,n) = \left((1-\beta)\cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}}\right)^{\frac{\rho}{\rho-1}} \tag{1}$$

  where $\rho > 1$ so that $u(c,0)$ is well-defined
- Homogeneous wage within gender: $w^{♂}$ and $w^{♀}$
- Gender wage gap: $\Gamma^w = w^{♂}/w^{♀}$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{♂,s} = u(w^{♂}, 0) \tag{2}$$

- Once married, husbands transfer α share of income to wives

$$V^{♂,m} = \lambda \cdot u((1-\alpha)w^{♂}, n^{m}) \tag{3}$$

  → $\alpha \in (-w^{♀}/w^{♂}, 1)$ is endogenous in equilibrium

  → $\lambda > 0$ is the common benefit of marriage (norms, companionship)

Marriage and Fertility – Women

- Single women solve

$$V^{\female,s} = \max_{c,l,n} \quad u(c,n) \quad \text{s.t.} \quad c = w^{\female} l, \quad l = 1 - \chi n \tag{4}$$

- Married women solve

$$V^{\female,m} = \max_{c,l,n} \quad \lambda \cdot u(c,n) \quad \text{s.t.} \quad c = \alpha w^{\male} + w^{\female} l, \quad l = 1 - \chi n \tag{5}$$

- Trade-off between consumption and fertility through female labor supply
- In the baseline, wives bear all childcare ($\chi^{\male} = 0$)

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with shape θ
- Marriage rates:

$$\mathcal{M}^g = \frac{(V^{g,m})^\theta}{(V^{g,m})^\theta + (V^{g,s})^\theta} \tag{6}$$

- Equilibrium: $\mathcal{M}^{\male} = \mathcal{M}^{\female} = \mathcal{M} \iff \frac{V^{\male,m}}{V^{\male,s}} = \frac{V^{\female,m}}{V^{\female,s}}$
- Two-dimensional price (α, n^m): marriage bundles income transfer *and* marital fertility

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- Two-dimensional price (α, n^m): marriage bundles income transfer *and* marital fertility
- Lemma 1: Unique interior equilibrium; α^* and n^m independent of λ, θ
- Lemma 2: Gains from marriage: $V^{\female,m}/V^{\female,s} = \lambda \cdot (1 + \alpha^* \Gamma^w)$, $n^m/n^s = 1 + \alpha^* \Gamma^w > 1$

Three-Way Trade-Off: Intuition

- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{[\lambda \cdot (1 + \alpha^* \Gamma^w)]^\theta}{1 + [\lambda \cdot (1 + \alpha^* \Gamma^w)]^\theta} \tag{7}$$

$$\Gamma^y = \Gamma^w / l^{♀} \tag{8}$$

$$l^{♀} = 1 - \chi n \tag{9}$$

- Insight: Higher n reduces $l^{♀}$, requiring low Γ^w for gender equality—but low Γ^w reduces the income-sharing attraction of marriage ($\alpha^* \Gamma^w$), lowering $\mathcal{M}$

Why Tensions Differ: The Wage Gap

Lemma 3: Γ^w determines which pairwise tension dominates

High Γ^w (Non-OECD)

(i) F–G amplification *strong*: $\Gamma^y = \frac{\Gamma^w}{1-\chi n}$, even moderate n pushes Γ^y above equality

(ii) F–M coupling *tight*: large marital fertility premium $q \gg 1$ ties marriage and fertility together

$\Rightarrow$ F–G tension dominates

Low Γ^w (OECD)

(i) F–G amplification *weak*: moderate n compatible with equality

(ii) F–M coupling *loose*: $q \approx 1$, so marriage and fertility decouple across countries

$\Rightarrow$ F–M tension dominates

Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	$+$	$+$	$+$ (widens)
Gender equality ($\uparrow w^{♀}$)	$-$	$-$	$-$ (narrows)
Reduce child costs ($\downarrow \chi$)	$+$	$+$	$+$ (widens)

- No single *level* policy can improve all three outcomes

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- No single *level* policy can improve all three outcomes
- Proposition: Combining $\uparrow w^{♀}$ with $\uparrow \lambda$ can improve all three, but requires delicate balancing between instruments
- Question: Is there a simpler single-instrument alternative?

Redistributing Childcare

Model with Husband-Provided Childcare

- Extension: $\chi^{♀}$ for wives, $\chi^{♂}$ for husbands, with $\chi^{♀} + \chi^{♂} = \chi$
- Paternal care efficiency $\eta \in (0, 1]$: father's raw time per child $= \chi^{♂}/\eta$
  $\rightarrow$ $\eta = 1$: equal efficiency; $\eta < 1$: father needs more time per unit of care
- Wife's effective childcare cost:

$$\tilde{\chi}^{♀} = \frac{w^{♀}\chi^{♀} + \alpha w^{♂}\chi^{♂}/\eta}{w^{♀} + \alpha w^{♂}} \tag{10}$$

- Income-weighted average: a child uses some of her earning capacity and some of his (deflated by η)
- Consider shifting δ effective units of childcare from wife to husband

Conditions for Joint Improvement

Proposition: Redistribution raises n, $\mathcal{M}$, and narrows Γ^y if:

1. Wife's effective cost falls: $w^{♀} > \alpha w^{♂}/\eta$

   Her freed earnings exceed the transfer-base loss from his care
2. Income convergence after fertility adjusts:

   Childcare redistribution more than compensates for additional care from higher n
3. Husband gains from adjusted contract:

   Despite more caregiving, the additional fertility is valuable enough

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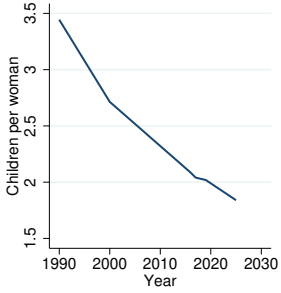
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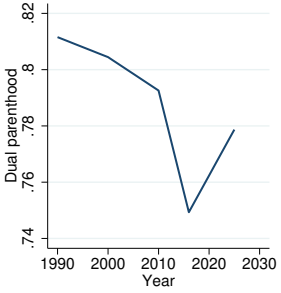
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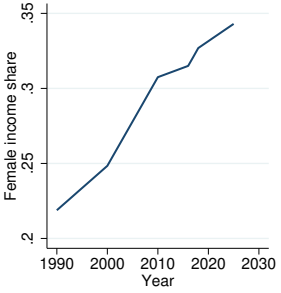
- All three conditions tighten as η falls $\implies$ minimum threshold $\bar{\eta}$
- Exists analytical ceiling on paternal share: $\chi^{\male}/\chi < \bar{\omega} < 1$

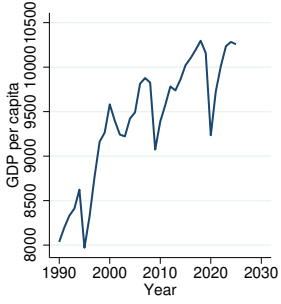
Quantitative Analysis

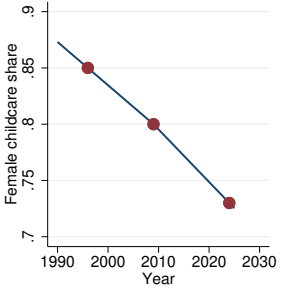
Mexico's Transition (1990–2025)









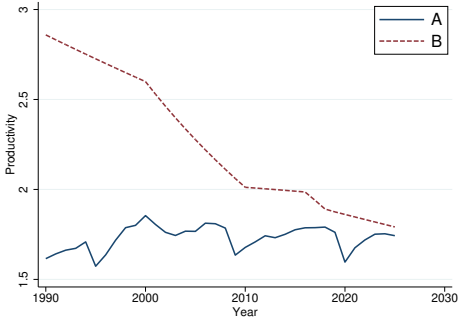


Calibration

model fit | validation

- Three time-varying inputs:
  - → $A_t = w_t^{♂}$: gender-neutral productivity
  - → $B_t = w_t^{♂}/w_t^{♀}$: relative wage gap
  - → s_t: female childcare share (ENUT 1996, 2009, 2024)
- Fix $\chi = 0.075$, $\eta = 1$. Calibrate 4 parameters to fertility & dual parenthood:

	Value	Target
β	0.068	Fertility level
ρ	3.0	Fertility trend
λ	0.945	Dual-parenthood level
θ	18.0	Dual-parenthood trend



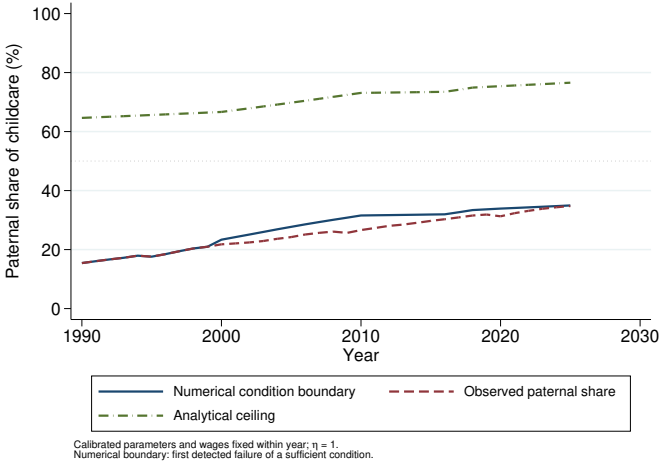
Relative Welfare

Year	CE: Single ♀	CE: Married ♀	CE: Single ♂
1990	2.20	2.15	0.06
1995	2.03	2.00	0.06
2000	1.74	1.80	0.03
2005	1.36	1.41	0.02
2010	1.06	1.10	0.02
2015	1.02	1.08	0.01

- Benchmarked to married male (CE $= 0$). Positive $=$ worse off
- Married-female CE halves from 2.15 to 1.08 while transfer shares fall
- All agent types converge toward married-male welfare

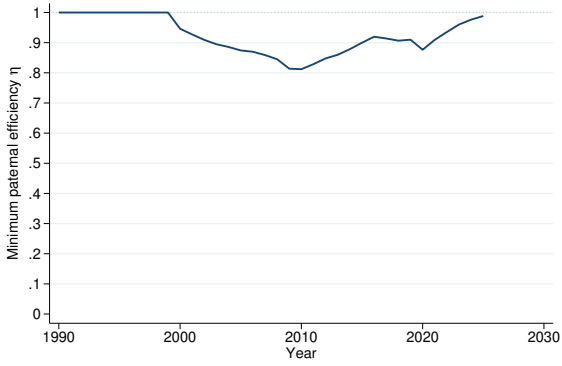
How Far Can Redistribution Proceed?

conditions



- All three conditions hold after 2000 at observed allocation
- Scope for redistribution falls short of equal sharing (condition 3)

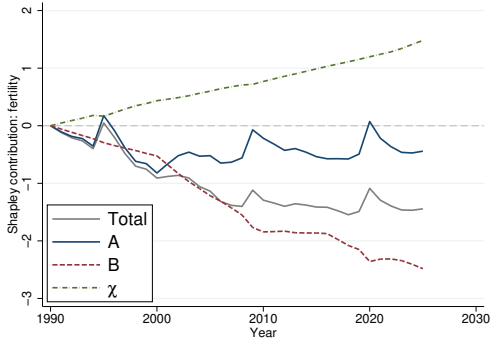
Relative Childcare Efficiency Thresholds

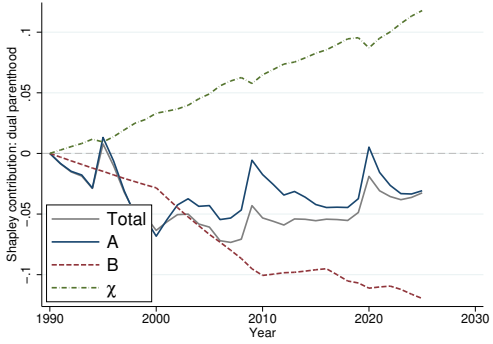


- $\bar{\eta} = 1$ in 1990–1999: conditions fail even with perfectly efficient fathers
- Declines to 0.81 by 2010: fathers could be 19% less efficient and redistribution still improves all three outcomes

Shapley Decomposition

full table





- Declining gender wage gap B_t is the dominant driver: 172% of fertility decline, 94% of income convergence, 96–114% of welfare convergence
- More equal childcare χ partially buffers fertility and dual parenthood

Conclusion

dynamic

- Three-way trade-off in with different dominant tension across countries
- Equilibrium marriage-market model offers two resolutions
  1. coordinated instruments
  2. childcare redistribution with bilateral conditions
- Mexico calibration
  - → declining wage gap drives the transition
  - → more equal childcare partially buffers fertility and marriage
  - → scope for further redistribution bounded below equal sharing

Bootstrap Inference

back

	Full		OECD		Non-OECD	
	Est.	SE	Est.	SE	Est.	SE
Trinity share	0.028	0.014	0.060	0.020	0.061	0.023
Shortfall (pp)	-9.65	1.40	-6.34	2.04	-6.38	2.30
$I(F;M)$ (bits)	0.121	0.054	0.076	0.054	0.014	0.025
$I(F;G)$ (bits)	0.239	0.066	0.000	0.017	0.060	0.044
$I(M;G)$ (bits)	0.015	0.018	0.038	0.037	0.027	0.033

1,000 replications clustered by country.

Decomposition Bootstrap

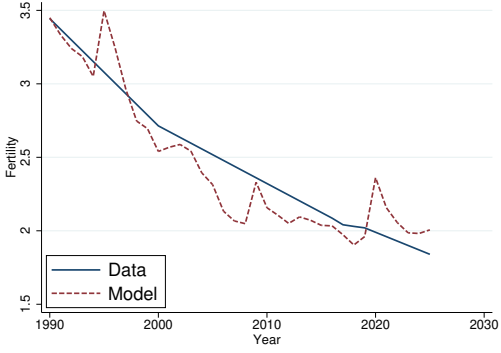
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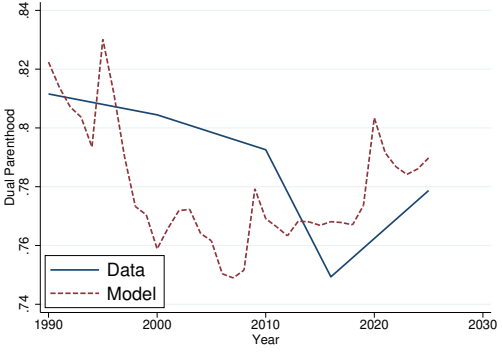
	OECD		Non-OECD	
Component	Point (pp)	p-value	Point (pp)	p-value
F−M tension	-4.02	0.003	-1.71	0.253
F−G tension	$+0.07$	0.964	-3.57	0.005
M−G tension	-2.86	0.044	-2.42	0.105
Three-way	$+0.47$	0.706	$+1.31$	0.300
Total	-6.34		-6.38	

- F−M tension: significantly negative in OECD ($p = 0.003$)
- F−G tension: significantly negative in non-OECD ($p = 0.005$), ≈ 0 in OECD ($p = 0.964$)

Model Fit: Fertility and Dual-Parenthood

back

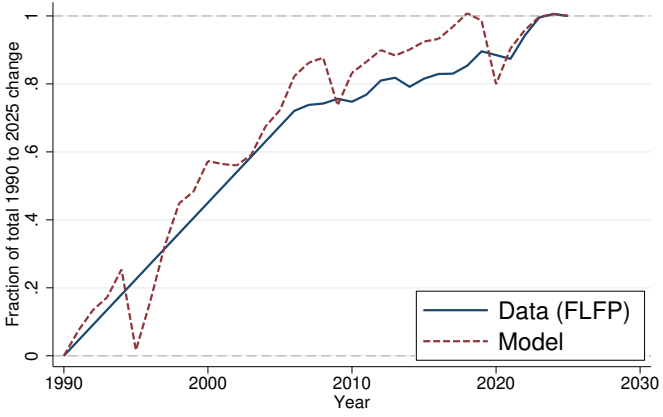




- Fertility: correlation 0.93, RMSE 0.18 children per woman
- Dual parenthood: correlation 0.29, RMSE 0.03; captures level and broad downward movement

Non-Targeted: Female Labor Supply

back



- Normalized to share of total 1990–2025 change; correlation $= 0.976$
- Reproduces rapid increase through 2008 and subsequent deceleration

Shapley Shares at 2025

back

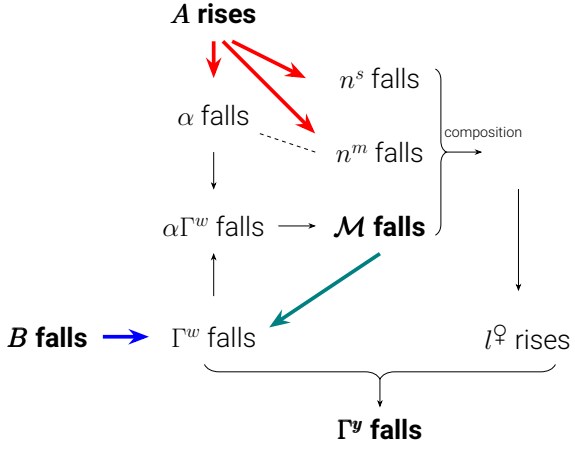
Outcome	A (%)	B (%)	χ (%)
Fertility	30.6	172.0	-102.6
Dual parenthood	94.5	367.6	-362.1
Gender income gap	3.9	94.4	1.7
CE: Single female	3.0	95.5	1.5
CE: Married female	2.4	113.6	-16.1
CE: Single male	22.4	69.5	8.1

% of total change from 1990 baseline. Negative share $=$ force moves outcome in opposite direction.

Dynamic Propagation

back

- Boys more sensitive to father absence (Autor et al. 2019, Wasserman 2020). Law of motion: $\Gamma^w_{t+1} = \mathcal{H}(\mathcal{M}_t), \quad \mathcal{H}' > 0$



Red: A_t rises Blue: B_t falls Teal: $\mathcal{M}_t \to \Gamma^w_{t+1}$ feedback